Maria Jose Usma

Robotics Perception /
3D Scene Understanding

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PROFESSIONAL SUMMARY

Master's student in Electronic Engineering focused on robotics perception, 3D scene understanding, and semantic mapping. Currently developing an object-level semantic Gaussian mapping pipeline that combines SplaTAM-based RGB-D reconstruction, SAM2 temporal mask propagation, multi-view CLIP semantic labeling, and KD-tree association with Gaussian primitives. Interested in semantic SLAM, visual-language mapping, and object-aware scene representations for robot navigation and embodied AI.

EDUCATION

Jeonbuk National University — Jeonju, South Korea
Master's Degree in Electronic Engineering

2024/09 - 2026/08

AI Robotics Laboratory.

National University of Colombia — Colombia
Bachelor's Degree in Electronic Engineering

2016/02 - 2021/12

Graduated with the second-highest GPA in the cohort.

PROJECTS

Object-Level Semantic Gaussian Mapping with Temporally Consistent SAM2 Masks and SplaTAM — *Master's Thesis (Ongoing)*

- Developed a modular pipeline for object-level semantic mapping on Gaussian Splatting volumetric representation using RGB-D frames, depth maps and camera poses, SAM2 masks and SplaTAM information.
- Used SAM2 for temporal object mask propagation and consistent 2D object IDs across video frames.
- Lifted 2D object masks into 3D by back-projecting masked depth pixels and transforming them into the world coordinate frame.
- Fused multi-frame object observations into object-level 3D point clouds and applied voxel downsampling, statistical outlier removal, and DBSCAN clustering for noise reduction.
- Performed open-vocabulary semantic labeling using multi-view

SKILLS

Programming: Python, C/C++

Robotics / 3D Vision: RGB-D processing, point clouds, camera poses, 2D–3D projection, semantic mapping, Gaussian Splatting

ML / VLMs: PyTorch, SAM2, CLIP, vision-language models, semantic segmentation

Tools: Open3D, Git, Linux, Docker, Qt, Pandas.

Methods: DBSCAN, KD-tree search, voxel downsampling, outlier removal, multi-view feature aggregation

AWARDS

GKS Scholarship 2023 - 2026
Global Korea Scholarship awarded by the Korean Government.

LANGUAGES

Spanish: Native
English: Fluent
Korean: Intermediate (TOPIK II - level 3, 2024)

CERTIFICATES

DS4A / Colombia 6.0: Graduated with Honors, Correlation One 2022

CLIP features extracted from selected object crops.
- Associated reconstructed object identities and semantic labels with Gaussian primitives using KD-tree radius search for object-aware Gaussian map visualization.
- Designed evaluation metrics for temporal mask consistency, 2D–3D geometric consistency, fusion quality, semantic confidence, and Gaussian-object association.

Data Science for All (DS4A) — *Ministry of Information and Communications Technologies of Colombia (MinTIC) and Correlation One (2022)*

- Selected as one of 1,800 participants from 15,000+ applicants for a national data science program.
- Worked with a team and government agency dataset to develop an algorithm for entity resolution and record matching under inconsistent and incomplete data.

EXPERIENCE

Kubo S.A.S, Colombia — *Web Developer*

2022/10 - 2023/07

- Developed and maintained web applications using frontend frameworks and REST APIs.
- Implemented features and supported maintenance tasks.

Titoma, Colombia— *Firmware Developer*

2020/11 - 2022/05

- Developed firmware in embedded C for ARM-based microcontrollers (STM32, ESP32, PIC).
- Implemented communication protocols (UART, SPI, I2C, Wi-Fi, Bluetooth).
- Worked with FreeRTOS, peripheral drivers, and real-time constraints.
- Built internal debugging and testing tools using Qt/C++ to support firmware development and hardware validation.
- Performed code reviews, testing, and technical documentation.

Titoma, Taiwan— *Embedded Systems Intern*

2020/02 - 2020/10

- Supported code development and version control using Git.
- Supported PCB design, component selection, and board debugging.

INTERESTS

- 3D Vision for Robotics
- Vision-Language Models
- Semantic Segmentation
- Scene Understanding
- SLAM